

Shantanu Vichare

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PROFESSIONAL SUMMARY

Lead QA engineer with ~11 years across SaaS quality, test automation, and applied AI. Own quality strategy for PowerSchool's Schoology cloud platform - zero P0 incidents over 4+ years - and ship production AI agent systems (GitHub Copilot Background Agents orchestrating Playwright, Browser, and Jira MCPs) that cut manual QA effort by 80%. Champion self-healing automation, cloud-native testing, and DevSecOps (OWASP ZAP, Burp Suite, SAST, DAST) in CI/CD. MSc AI and ML in progress, Liverpool John Moores University.

CORE COMPETENCIES

AI Quality Engineering	Agent-Based Test Automation	Self-Healing Test Frameworks	Cloud-Native Testing Strategies
DevSecOps in CI/CD	Test Strategy and Mentorship	Playwright / WebdriverIO / API Testing	Quality Observability and Dashboards

WORK EXPERIENCE

PowerSchool

2020 - Present

Lead Quality Assurance Engineer - Bengaluru, India

- Scope expansion:** Promoted into AI-enabled Quality Engineering leadership in 2023 after 3 years leading Core QA; now own AI test tooling strategy for Schoology while mentoring 20+ engineers across 6 Agile squads.
- Agentic QA ecosystem:** Architected a multi-agent test platform with Playwright, Browser, and Jira MCPs orchestrated by GitHub Copilot Background Agents and LLMs (Claude, OpenAI, Gemini) for autonomous test authoring, execution, and ticket triage across Schoology.
- Self-healing AI tests:** Engineered an AI Context Graph anchoring LLM agents to live OpenAPI contracts and canonical suites, so Playwright tests auto-repair drifted locators on UI change with 90%+ first-pass accuracy and zero hallucinated steps.
- Autonomous API test synthesis:** Designed an agent-driven pipeline that generates, validates, and runs Playwright API tests from OpenAPI specs - retiring manual Postman workflows and lifting regression coverage from 55% to 95%.
- 80% QA effort reduction:** Scaled Cursor- and Copilot-driven agent workflows across the tribe, compressing prompts-per-task from 15 to 2-3 and reclaiming 4+ engineer-weeks per release cycle.
- Cloud-native test strategy:** Orchestrated containerized Playwright + WebdriverIO runners across ephemeral cloud environments with parallel workers, cutting regression runtime by 70% and unlocking same-day production deployments.
- DevSecOps in CI/CD:** Embedded OWASP ZAP DAST, Burp Suite checks, dependency audits, and policy gates into Bamboo, GitLab, and GitHub Actions pipelines - blocking vulnerable code at merge time.
- Zero-incident SaaS quality (4+ years):** Directed QA strategy for Schoology, sustaining zero P0 production incidents while delivering 20+ complex epics on schedule.
- Quality observability cockpit:** Shipped a MERN-stack internal release tool unifying CI, defect, and test-coverage telemetry through live dashboards - compressing release checklist time from ~4 hours to ~15 minutes.

FireEye (Mandiant)

2018 - 2020

Software Development Engineer in Test - Bengaluru, India

- Security product test frameworks:** Developed UI and security testing frameworks from scratch for Linux, Windows, and Mac endpoint security products, cutting release escape defects ~35% release-over-release.
- E2E automation at scale:** Composed Python + Nosetests suites validating detection and prevention across 8+ security releases, shortening regression cycles ~40%.
- Cross-functional quality:** Partnered with developers and product managers across 5+ distributed teams to triage and resolve complex security software bugs.
- Methodology uplift:** Spread Python, Selenium, and Robot Framework practices, raising the baseline quality bar for 15+ incoming engineers.

Quick Heal Technologies

2015 - 2018

Associate Threat Research Engineer - Pune, India

- Automation-first malware triage:** Engineered internal tooling that replaced manual malware analysis workflows, reclaiming ~12 analyst-hours per week for higher-value threat research.
- Incident response and RCA:** Led customer escalation and root-cause analysis across 50+ security incidents, feeding remediation insights back into the product detection roadmap.
- Product security features:** Proposed 2 threat-detection capabilities shipped into Quick Heal products, strengthening endpoint protection for millions of users.

SELECTED PROJECTS

AI Context Graph Architecture

Developed an internal grounding framework anchoring LLM test generators to live API contracts and canonical suites - the reusable backbone of the self-healing QA platform at PowerSchool, adopted by 3+ product teams.

Specialist QA Agent Suite

Built specialist downstream agents for accessibility (WCAG 2.1), internationalization (i18n), and security NFR validation plugged into the multi-agent QA platform - automating compliance checks that previously required dedicated manual specialist review.

DevSecOps Pipeline Blueprint

Delivered a reusable CI/CD security blueprint combining OWASP ZAP, Burp Suite, SAST, and dependency audits as gated stages in Bamboo, GitLab, and GitHub Actions - catching 30+ vulnerabilities pre-production.

EDUCATION

MSc, Machine Learning and Artificial Intelligence - Liverpool John Moores University, England	2025 - 2027
Post Graduate Diploma, AI and ML - IIIT Bangalore	2025 - 2026
Bachelor of Engineering, Information Technology - Shah and Anchor Kutchhi Engineering College, Mumbai	2012 - 2015

CERTIFICATIONS

CopilotXcelerate: AI-Powered Python for Data Science - Upgrad	2023
Certified Ethical Hacker (CEH) - Simplilearn	2019
Oracle Certified Expert, Database SQL - Oracle	2016

SKILLS

AI and Agent Tooling: Cursor, Claude Code, Playwright MCP, Browser MCP, Jira MCP, GitHub Copilot Background Agents, VS Code Copilot, RAG, Multi-Agent Orchestration, LLMs (Claude / OpenAI / Gemini), AI Context Graphs, Self-Healing Test Generation

Test Automation: Playwright, Selenium, WebdriverIO, Robot Framework, PHP Codeception, Nosetests, BDD, API Testing, Visual Regression Testing

Cloud and CI/CD: Bamboo, GitLab, GitHub Actions, Containerized Test Runners, Parallel Workers, Ephemeral Environments, Cloud-Native SaaS Testing

Security and Performance: OWASP ZAP, Burp Suite, OWASP Top 10, DevSecOps, SAST, DAST, Dependency Scanning, JMeter, LoadRunner, Penetration Testing

Engineering: Python, TypeScript, JavaScript, PHP, MERN Stack

Process: Agile and Scrum, Test Strategy and Planning, Mentorship, Risk Assessment, Vendor Management, Compliance

Languages: English, Hindi, Marathi